

Robot Operating System (ROS) - Intro

Intelligent Robotics I

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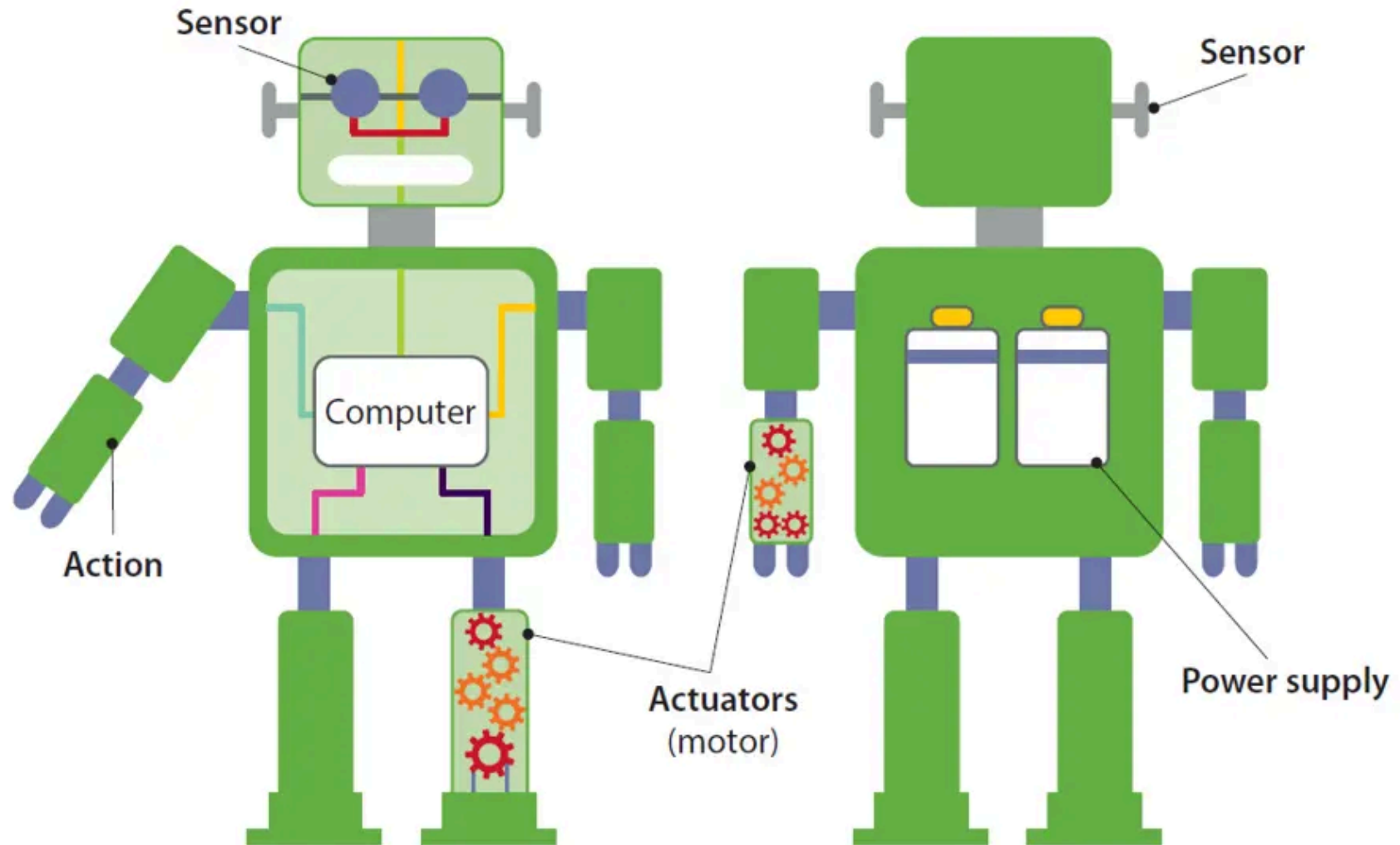
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What is ROS2?

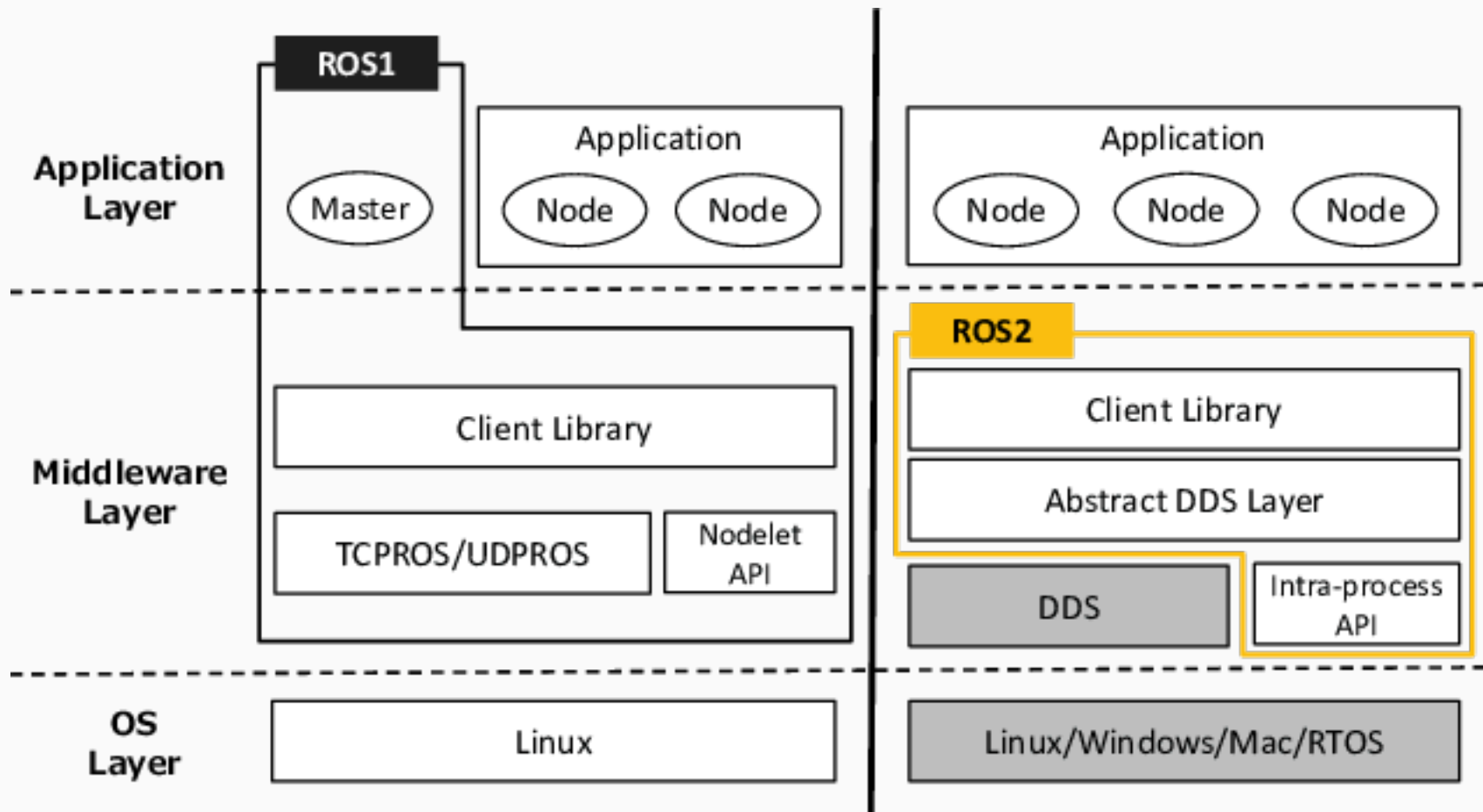
Robot Parts



Source

1. middleware robot parts
2. package management
3. hardware abstraction

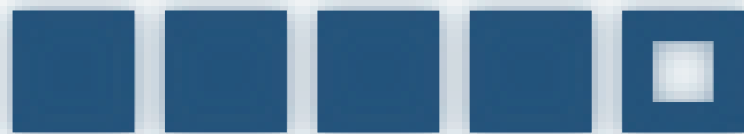
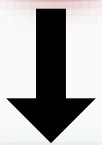
Middleware: responsible of message passing in distributed systems



Source

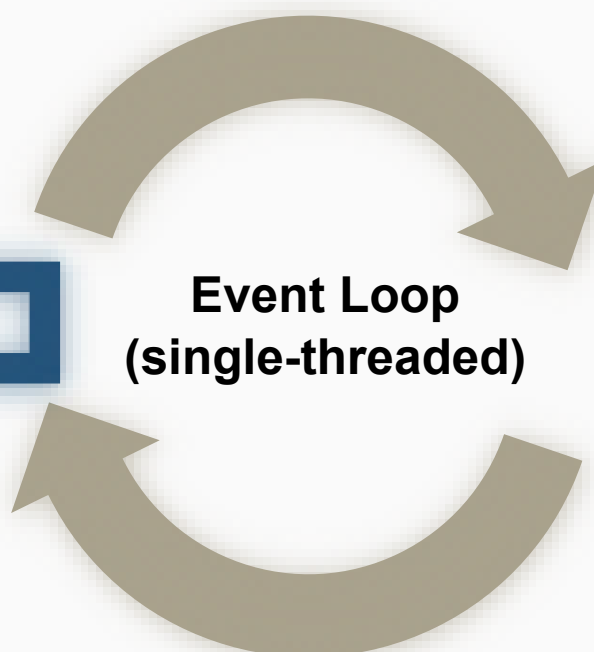
- pooling: constant refresh zone memory, not efficient
- event-driven: interrupts, needs communication

Event Emitters



Event Queue

Event Loop (single-threaded)



Event Handler

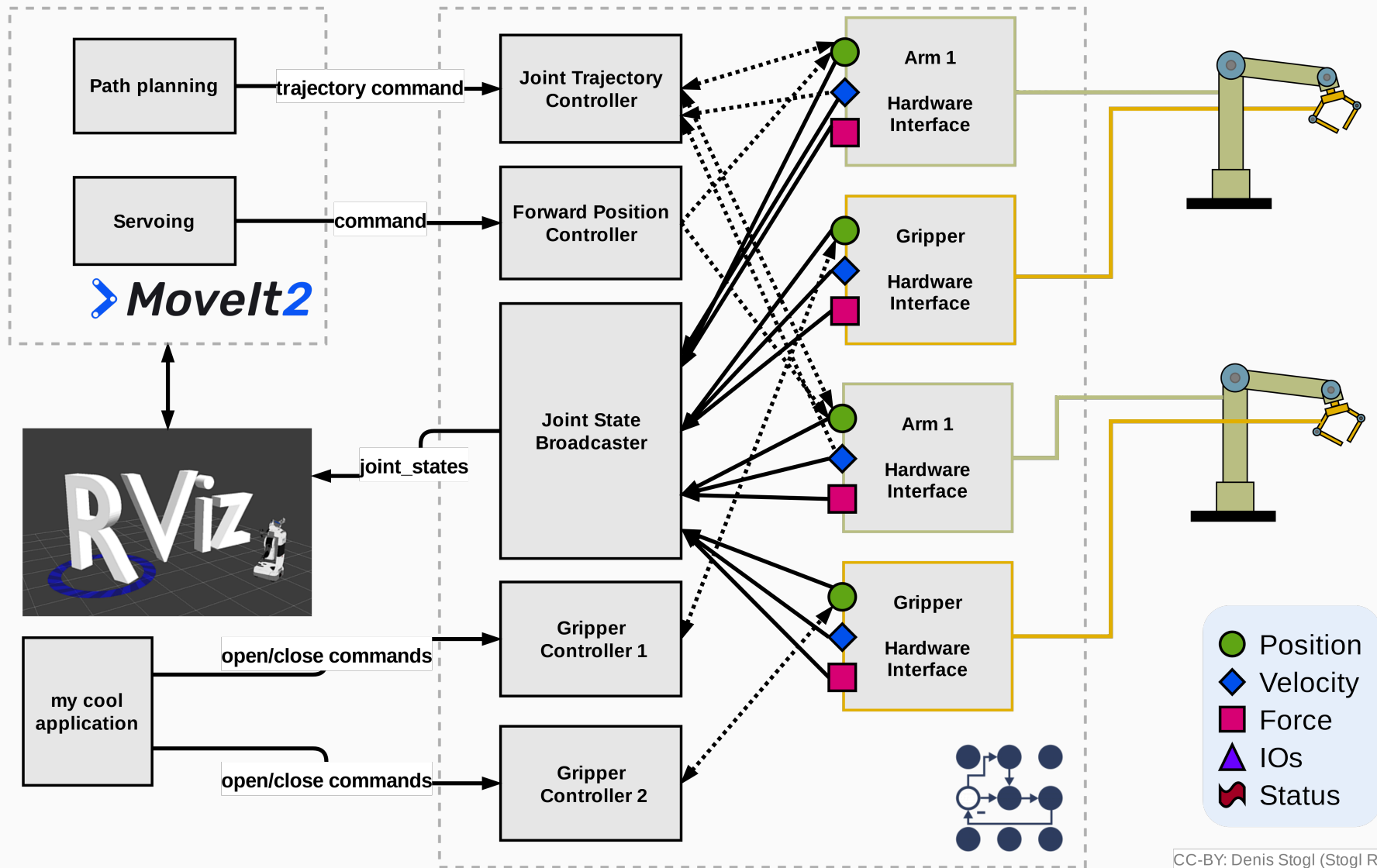


States

Package = individual module

Package	Short Description
rclpy	Python API
tf2	coordinate transformations
nav2	navigation system for robots
rviz2	3D visualization tool
moveit2	motion planning for robotic arms
gazebo	physics simulation

Hardware Abstraction



Source

Concept	Description
nodes	Programs that perform tasks in ROS.
packages	Collections of related nodes and libraries.
topics	Channels for sending messages between nodes.
services	Allow request-response communication between nodes.
actions	Handle long-running tasks.

ROS2



Timeline



Source

Packages Available



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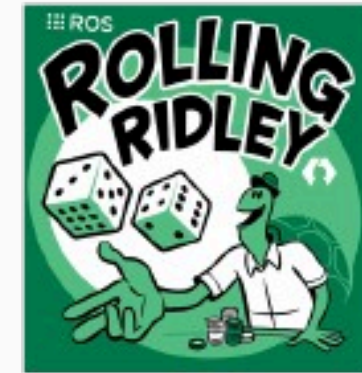
1732



1346



1384



1251

Source

ROS1	ROS2
Ubuntu	Ubuntu, OS X, Windows 10
C++03	C++11, C++14
TCP	Data Distribution Service (DDS)
central master node	distributed
cmake	cmake, python, ...
third party real time	supports real-time nodes

Resources

- “ROS2 Documentation Humble” by Open Robotics / [link](#)
- “ROS2 Tutorials” / [link](#)
- “Free Robotics Crash Course” by Robotics Back-End / [link](#)
- “Get Ready to Build Robots with ROS!” by Articulated Robotics / [link](#)